

# P3 CAPITAL

## CAP Travel Assistance Platform

*Business Requirements Document (BRD)*

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|------------------|--------------------------------------------------------------------------|
| Document Version | 1.0 (Initial Draft)                                                      |
| Prepared By      | Business Analysis Team                                                   |
| Date             | 2025                                                                     |
| Status           | Draft 1                                                                  |
| System Name      | CAP Travel Assistance Platform                                           |
| Technology Stack | React (Frontend)   .NET & Java (Backend)   Stripe (Payments)   AWS/Cloud |

# 1. Executive Summary

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The CAP Travel Assistance Platform is a comprehensive, multi-tenant travel assistance and insurance management system. The platform serves multiple user types across distinct portals: Agents (B2B), Travelers/Consumers (B2C), Enterprise Resellers (B2B2C), and Hybrid/Admin Dashboard users. It enables the purchase, management, and administration of travel assistance plans — with a multi-step checkout wizard, policy and order management, Stripe payment processing, travel risk intelligence, and third-party integrations including Cigna and Google Maps.

This Business Requirements Document captures functional requirements derived from the analysis of 60+ React UI components, their workflows, API endpoints, and business logic. The system architecture reflects a modern, scalable SaaS application with role-based access, Redux state management, and Ant Design component library.

The platform is organized across four major functional areas:

- **Agent & Consumer Portal** — plan purchase wizard, agent order management, authentication flows, Cigna integration, family purchase, and plan upgrades
- **B2B2C Partner Portal** — white-label branded experience for reseller organizations
- **Hybrid Admin Dashboard** — B2B2C organization management with interactive travel risk map, user/agent/traveler lifecycle management, audit logs, notifications, and email template customization
- **Admin Reporting Module** — full business intelligence suite covering invoices, billing, traveler tracking reports, airport/flight/hotel/car reports, and organizational license management

## 2. Project Overview

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### 2.1 Background

The system was analyzed from the React frontend application across three code review iterations. Backend services (.NET, Java) were not directly executed due to local database connectivity limitations; however, API contracts and business logic were inferred from frontend integration points.

### 2.2 Objectives

- Provide a comprehensive travel assistance plan purchase and management platform
- Support multiple sales channels: Agent (B2B), Consumer Direct (B2C), and Reseller (B2B2C)
- Enable seamless multi-step plan purchase with real-time pricing and discount management
- Provide a Hybrid Admin Dashboard for B2B2C organization management and travel risk monitoring
- Provide a full Admin Reporting Suite for business intelligence, invoicing, and travel tracking
- Integrate with Stripe for secure payment processing and Google Maps for travel visualization
- Support Cigna partner integration for user provisioning and access management

### 2.3 Scope

#### In Scope

- Agent portal: login, order management, profile management, password management
- Consumer (B2C) portal: multi-step plan purchase wizard, quote-initiated flow
- Reseller (B2B2C) portal: branded white-label experience with custom theming
- Hybrid Admin Dashboard: travel map, organization management, users, agents, travelers, divisions, audit logs, notifications, email templates, country risk intelligence
- Admin Reporting Module: invoices, billing, traveler itinerary viewers, 10+ report types, license management, products, roles
- Cigna partner integration: user provisioning, activation, rejection flows
- Family purchase flow with multi-traveler support
- Plan upgrade management
- Authentication: login, forgot password, reset password, change password
- Payment integration: Stripe checkout

#### Out of Scope

- Backend infrastructure setup and database migration
- Third-party travel insurance claims processing
- Mobile native applications (iOS/Android)
- Offline functionality

### 3. Stakeholders & User Roles

| User Role                     | Description                                                                                                                           | Primary Portal              |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Agent                         | Licensed agents selling travel plans on behalf of their organization. Manage their own orders, commission data, and client purchases. | Agent Portal (/agent)       |
| Traveler / Consumer           | End customers purchasing travel assistance plans directly. Individual or family group.                                                | B2C Portal (/cap)           |
| Reseller (B2B2C)              | Enterprise resellers offering white-labeled CAP plans with custom branding.                                                           | Partner Portal (/partner)   |
| Hybrid Admin / Travel Manager | Administrators managing the B2B2C organization dashboard: users, agents, travelers, divisions, risk monitoring.                       | Hybrid Dashboard (/hybrid)  |
| Administrator / CSR           | Internal super-admins and customer service reps with full system access including all reporting.                                      | Admin Panel (/admin, /user) |
| Cigna Partner                 | Enterprise partner using Cigna integration for employee plan access provisioning.                                                     | Cigna Module (/cigna)       |
| System / API                  | Backend services, Stripe, Google Maps, RiskLine API, third-party integrations.                                                        | N/A                         |

## 4. Functional Requirements

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### 4.1 Authentication & Account Management

#### 4.1.1 User Login

Secure login page for all user roles with email/password fields, Remember Me toggle, Forgot Password link, and loading indicator during authentication.

- Accept email and password credentials
- Support 'Remember Me' functionality using localStorage persistence
- Redirect authenticated users to their role-appropriate dashboard
- Display loading spinner during authentication attempt
- Show clear error messages on authentication failure
- Auto-focus email input field on page load
- Clear sensitive data from localStorage after successful authentication

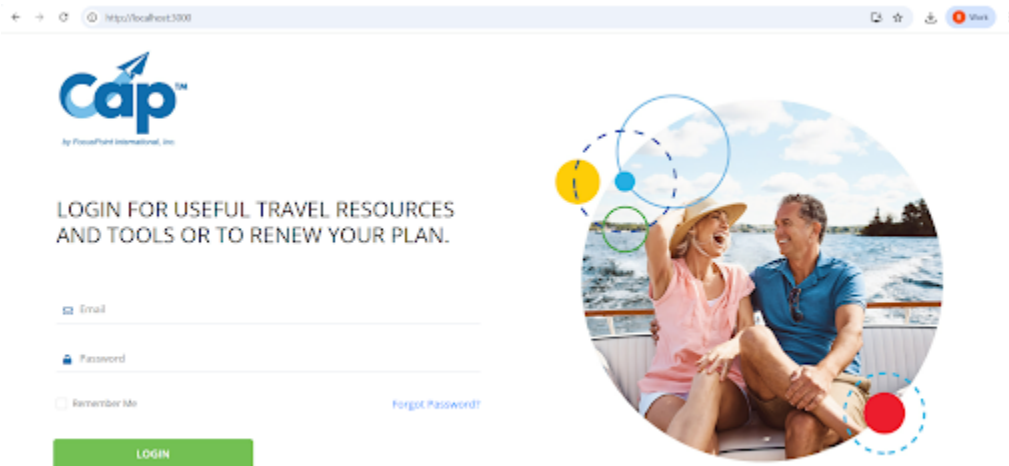


Figure: Login Screen

#### 4.1.2 Forgot Password

- Provide a self-service password reset request page accessible from the login screen
- Validate email format via regex before submission
- Call the forgotPassword API endpoint with the user's email
- Display success notification and redirect to login on successful submission
- Show a 'Please wait...' button state during processing to prevent duplicate submissions

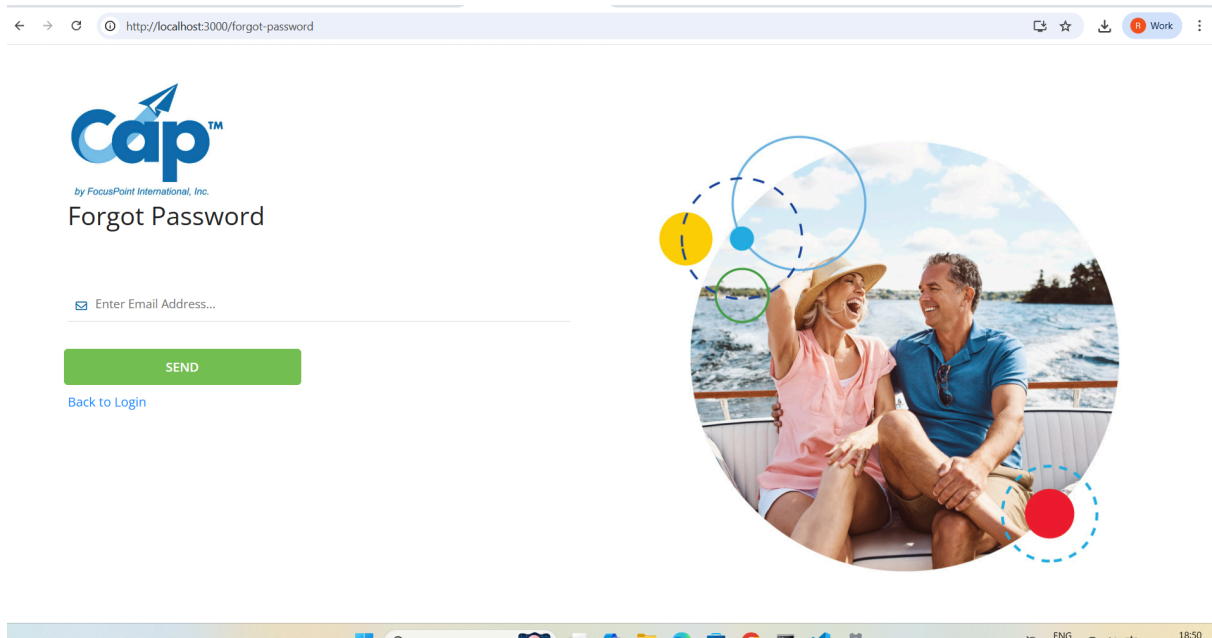


Figure: Forgot Password Screen

#### 4.1.3 Reset Password

- Accept a secure token-based reset link with username and token as URL query parameters
- Enforce strong password requirements: uppercase, lowercase, number, special character, minimum 8 characters
- Provide real-time password strength feedback via an interactive popover
- Validate that new password and confirm password fields match before submission
- On success, redirect to login page and show success notification
- Handle expired token errors with a specific error message

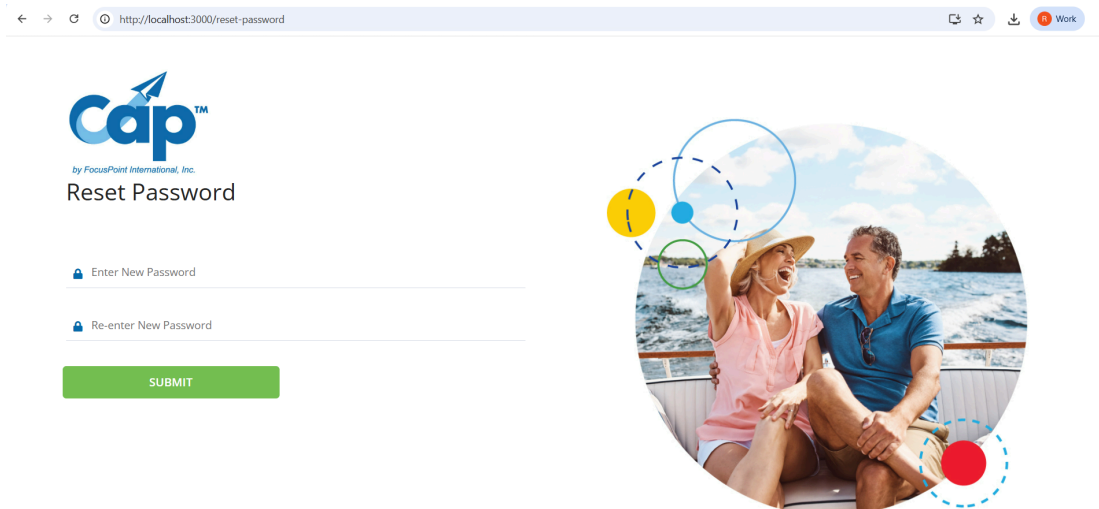


Figure: Reset Password Screen

#### 4.1.4 Change Password (Authenticated)

- Allow authenticated users to change their password from the profile area
- Require the current (old) password before allowing a new password
- Enforce the same strong password requirements as the Reset Password flow
- Detect unconfirmed accounts and display a forced password change warning
- On successful change, mark account as confirmed in localStorage and redirect to dashboard
- Show live validation popover with requirement status on each keystroke

The screenshot shows a web browser window with the URL `http://localhost:3000/agent/my-profile/change-password/`. The page is titled "Change Password" and features a dark blue sidebar on the left with the "cap Travel Assistance Portal" logo and navigation links: "Manage Orders", "Countries", "My Profile", and "Change Password" (which is highlighted). The main content area contains three password input fields, each with a red asterisk and a label: "Old Password:", "New Password:", and "Confirm Password:". Each field has a small eye icon to the right for toggling visibility. Below the fields is a blue "Update" button. The browser's top bar shows standard navigation icons and a "Work" tab.

Figure: Change Password Screen

## 4.2 Agent Portal

### 4.2.1 Agent Order History (Manage Orders)

The Agent Order History module provides agents with a full paginated view of all orders placed under their account.

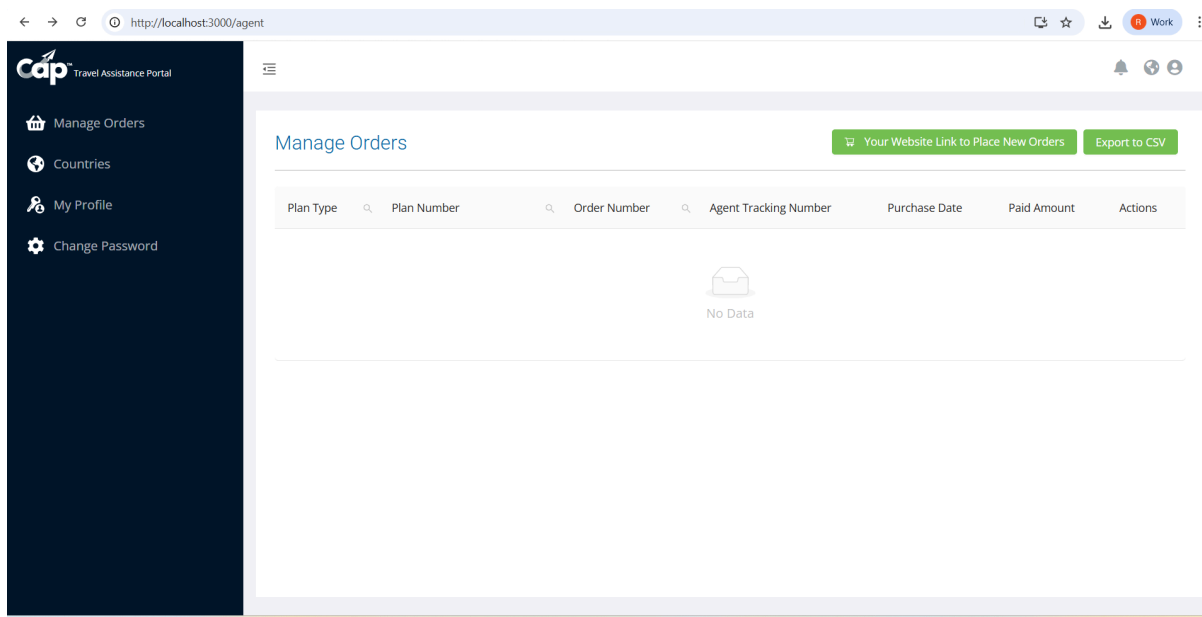


Figure: Agent Portal — Manage Orders Screen

- Display a paginated, sortable table of orders filtered by the agent's code
- Columns: Plan Type, Plan Number (Membership), Order Number, Agent Tracking Number, Purchase Date, Paid Amount, Actions (View)
- Support column-level search for: Plan Type, Membership Number, Order Number
- Format amounts as USD currency; display purchase dates in MM/DD/YYYY format
- Provide 'Export to CSV' functionality with All or custom Date Range options
- Show dynamically generated 'Your Website Link to Place New Orders' button

### 4.2.2 Agent Profile (My Profile)

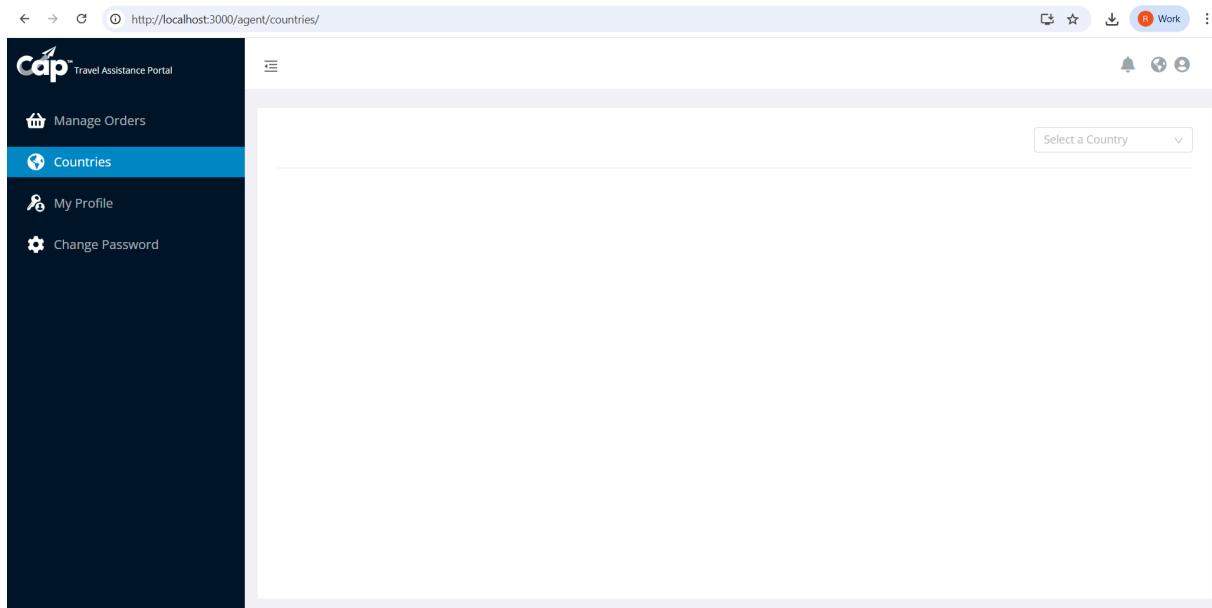
- Display and allow editing of: First Name, Last Name, Email, Date of Birth, Phone, Address, Country, State
- Validate required fields with regex patterns for names, email format, and phone format
- Disable email editing if the account was pre-provisioned (checkexisting flag)
- Provide a Travel Alerts toggle (conditional on user permissions)
- Update localStorage with new name and phone on successful profile save

### 4.2.3 Change Password

- Accessible from the left navigation; enforces strong password policies with real-time visual feedback
- Require the current (old) password before allowing a new password
- On success: mark account as confirmed, redirect to agent dashboard



## 4.3 Admin Module — Country Management



*Figure: Admin Module — Country Management*

- Display countries grouped by region in tab-based navigation
- Render each country with its flag image and name in a responsive grid layout
- Provide a global searchable dropdown to select any country across all regions
- Navigate to a country-specific detail/management page on selection

## 4.4 B2C Purchase Wizard

The B2C purchase flow consists of a 5-step wizard guiding consumers through plan selection to order confirmation. Each step enforces sequential progression; users cannot skip steps via direct URL access.

| Step   | Page                | Description                                                                                         |
|--------|---------------------|-----------------------------------------------------------------------------------------------------|
| Step 1 | Plan Type Selection | User selects a travel plan tier. Profile completeness and age (18+) validated. Plan saved to Redux. |
| Step 2 | Plan Term Selection | User selects coverage duration and destinations. Plan term details saved to Redux.                  |
| Step 3 | Personal Details    | User enters traveler registration details. URL query params decoded for pre-filled quote data.      |
| Step 4 | Billing Information | Pricing summary with promo discounts. Per-traveler plan selection. Stripe payment integration.      |
| Step 5 | Purchase Summary    | Final order review. User confirms and places order via placeOrder API. Redirects to thank-you.      |

### 4.4.1 Plan Selection (Step 1)

- Display available plan tiers with pricing as selectable cards
- Show 'best value' badge on plans with 'med' in the product name
- Validate profile completeness (contact number, home country) and minimum age 18 before allowing purchase
- Display profile validation modal redirecting to My Profile if validation fails
- Persist selected plan to Redux state (SavePlanType, SaveStep)

### 4.4.2 Personal Details (Step 3)

- Decode base64-encoded URL query parameters: start/end dates, travel countries, residence details, trip type, currency, traveler count
- Persist decoded parameters to localStorage for page refresh resilience
- Support annual plan toggle and dynamic traveler count adjustment
- Support currency switching (USD/CAD) with rate conversion
- Route guard: redirect to plan-term if step < 2

### 4.4.3 Billing Information (Step 4)

- Support per-traveler plan selection/upgrade
- Calculate total price dynamically: country tier + day package + traveler count + selected product
- Apply promotional code discounts (percentage or fixed amount)
- Integrate Stripe payment form via StripeProvider component
- Support both USD and CAD currency display
- Prevent accidental page unload (Beforeunload hook)

### 4.4.4 Purchase Summary (Step 5)

- Display final plan fee with all promotions and discounts applied
- Build and submit complete order payload via placeOrder API on confirmation
- Redirect to /thank-you on successful order placement

#### **4.4.5 CAP Dashboard / Route Redirect Handler**

- Convert legacy shortcut routes to current internal URLs using companyID
- For 'newplan' route: validate profile completeness and age  $\geq 18$
- For B2B2C org type: redirect to /partner/purchase/:organizationId

#### **4.4.6 Pricing Engine (PriceCalculation Component)**

- Calculate base price from selected country tiers and day package
- For US domestic plans: use state-level risk zone pricing (max risk zone across selected states)
- For international plans: use country-tier pricing (max tier among selected countries)
- Apply coupon discounts (percentage or fixed) and support CAD currency conversion (rate: 1.3644)
- Update automatically when plan term, traveler count, coupon, or currency changes

## 4.5 B2C Order Flow (Quote-Initiated)

- Accept quote parameters via base64-encoded URL query string from external quote builder
- Decode: start/end dates, travel type, countries, US states, residence details, trip type, currency
- Auto-set plan type from configuration and save to Redux
- Persist decoded parameters in localStorage for session resilience
- Route guard enforces step 3+ to access billing summary

## 4.6 Cigna Partner Integration

### 4.6.1 User Provisioning (Create Access)

- Admin form to create Cigna user access: First Name, Last Name, Email, Phone, Policy/Account Number
- Validate email uniqueness on blur; show modal if email already exists
- Validate policy/account number as 6 or 9 alphanumeric characters
- Call CreateUserAccess API on submission; reset form on success

### 4.6.2 User Activation

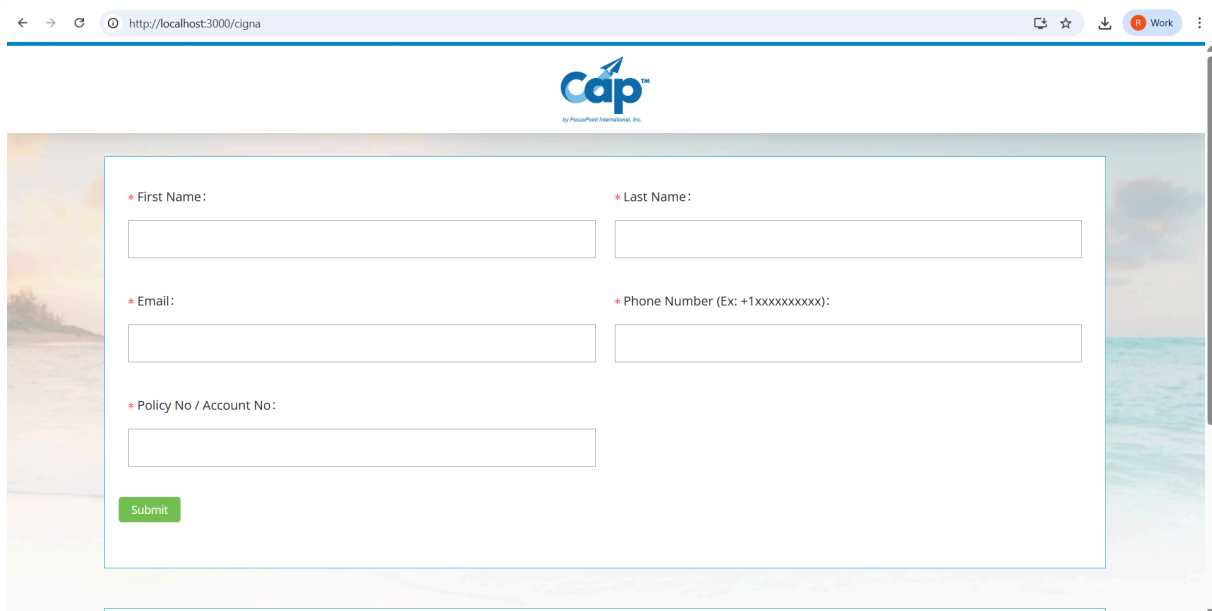
A screenshot of a web browser displaying a user activation form. The browser's address bar shows 'http://localhost:3000/cigna'. The page features the 'Cap' logo at the top center. The form is a white box with a light blue border, containing five input fields with red asterisks indicating required fields: 'First Name', 'Last Name', 'Email', 'Phone Number (Ex: +1xxxxxxxxx)', and 'Policy No / Account No:'. A green 'Submit' button is located at the bottom left of the form. The background of the page is a scenic image of a beach at sunset.

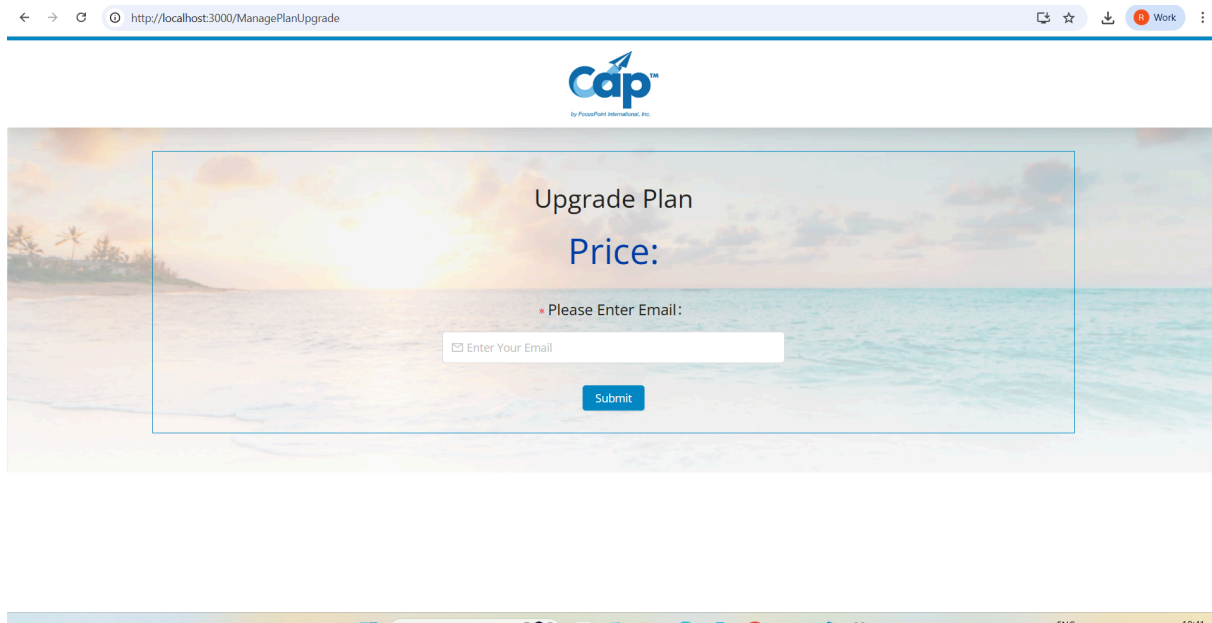
Figure: Cigna — Approve User Access Confirmation Page

- Token-based one-shot confirmation page at /cigna/approvecreateuser/:id
- Calls GET /api/approvecreateuser/:id and displays resulting status message

### 4.6.3 User Rejection

- Corresponding rejection page at /cigna/rejectcreateuser/:id
- Calls GET /api/rejectcreateuser/:id and displays status message

## 4.7 Plan Upgrade Management



*Figure: Plan Upgrade — Purchase Plan Screen*

- Allow administrators/agents to initiate a plan upgrade for a user by email
- Call B2BEmployeeUpgrade API to check user eligibility
- If eligible: display Stripe payment section and auto-scroll to it
- If not eligible: show modal with redirect to CAP website

## 4.8 Family Purchase Flow

### 4.8.1 Family Plan Purchase

- Allow purchase of a family plan with a primary member and multiple child travelers
- Validate parent email existence; validate child emails for uniqueness and employee status
- Support dynamic addition and removal of child travelers
- Recalculate total price dynamically as travelers are added/removed
- Fields per traveler: First Name, Last Name, DOB, Email, Country, Phone, Gender, Relationship
- Emergency contact fields per traveler: First Name, Last Name, Phone
- Disable Stripe payment section until form validation passes

### 4.8.2 Family Purchase Thank You

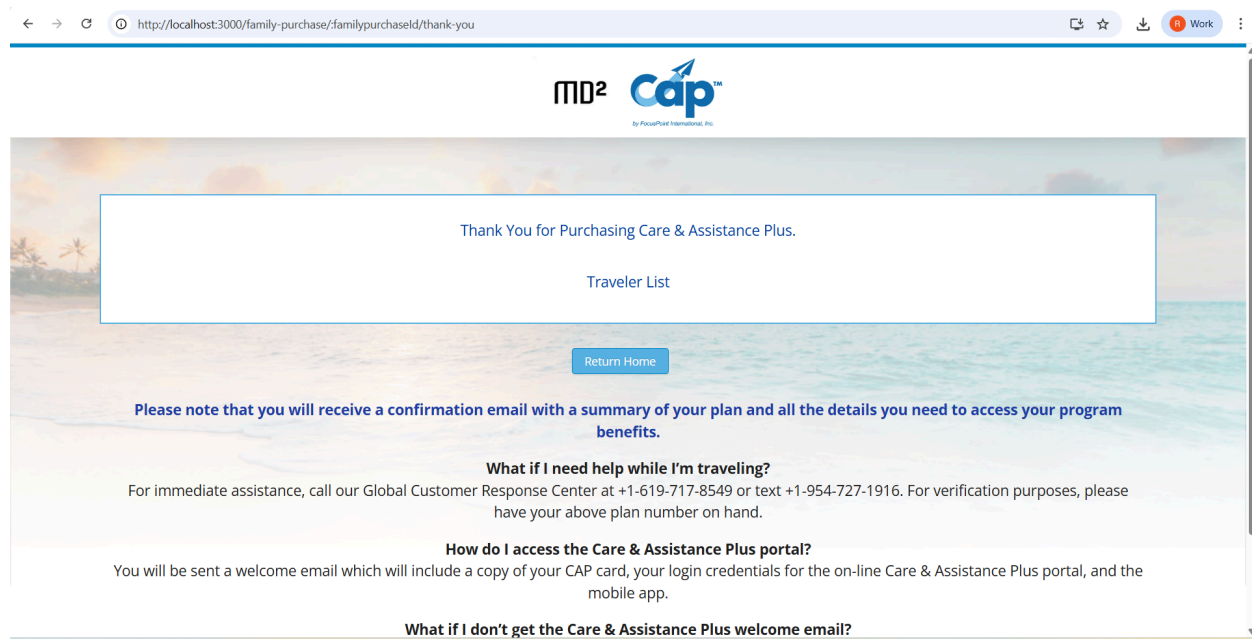


Figure: Family Purchase — Thank You / Confirmation Screen

- Display confirmation page after successful family plan purchase
- Read purchased traveler names from localStorage (childUserInfo)
- Provide next steps: confirmation email, assistance phone numbers, portal access, support contact

## 4.9 B2B2C Partner / Reseller Portal

- Apply dynamic theming from reseller settings: custom background image and primary color
- Override CSS for header, banners, buttons, and icons with the reseller's primary color
- Initialize Google Analytics (G-N02B8BDNND) and Microsoft Clarity on production environments
- Compose full layout: branded header, routed B2B2C content, and branded footer



## 4.10 Hybrid Admin Dashboard

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The Hybrid Admin Dashboard is a comprehensive B2B2C organization management portal for Travel Managers, Administrators, and CSRs. It provides travel risk monitoring on an interactive map, full user/agent/traveler lifecycle management, division management, audit logging, notifications, and customizable email templates.

### 4.10.1 Travel Map Dashboard (Interactive)

An interactive Google Maps-based dashboard providing a real-time global view of all travel activities, risks, and assistance needs for the organization.

- Display a world map (default zoom 3) with custom GeoJSON country boundary overlays
- Color-code countries by risk level: Low (green), Moderate (light green), Medium (yellow), High (orange), Extreme (red)
- Support multiple marker types simultaneously: Airport, Hotel, Assist Traveler, Traveler Location, Asset Tracking, and Alert markers
- Display flight routes as dashed red polylines when a traveler marker is clicked
- Provide collapsible filter panels for: Traveler Itineraries, Alerts, Risk Levels, Assist Travelers, Traveler Locations, and Asset Tracking
- Filter traveler itineraries by: date range, airline, airport, passenger name, flight number, PNR, and organization
- Filter alerts by: country, risk level, time duration (1d, 7d, 30d), and category
- Toggle individual risk level visibility to update map country coloring dynamically
- Allow Travel Managers (non-Cigna) to initiate assistance requests via modal for travelers with active memberships
- Fetch data on load: traveler itineraries, assist travelers list, RiskLine categories/alerts, country risk levels, traveler locations, asset tracking data
- Info windows on marker click show: itinerary details, hotel info, assistance request details, or alert information
- Restrict Assist button visibility to Travel Manager role only

### 4.10.2 Hybrid My Profile

User profile management for hybrid dashboard users, with emergency/alternate contact management and travel alert preferences.

- Display and allow editing of: First Name, Last Name, Email, Date of Birth, Phone, Address, Country, State
- Emergency Contact section for membership users; Alternate Contact section for non-membership users
- Emergency contact fields: First/Last Name, Relationship, Phone; Email field shown only for non-membership users
- Travel Alerts toggle visible only if user has 'update\_receive\_travel\_alerts' permission
- Update localStorage with new name/phone on successful save
- All text supports internationalization (IntlMessages)

### 4.10.3 Hybrid Change Password

- Same strong password policy as other portals (uppercase, lowercase, numeric, special char, 8+ chars)
- On success: mark account as confirmed in localStorage and redirect to hybrid dashboard
- Forced password change warning for unconfirmed accounts
- Real-time popover with per-requirement color-coded icons

### 4.10.4 Manage Orders (Reseller Order History)

Order history management for reseller/organization users to track sales and commissions.

- Display paginated table of orders: Product Name, Membership Number, Order Number, Actions
- Show commission totals in both USD and CAD currencies
- Column-specific search for Product Name, Membership Number, Order Number
- Date range filtering with predefined options: Today, Yesterday, This Week, Last Week, This Month, Last Month
- Export to CSV: All orders or custom date range via `/api/ExportOrders/{orgId}/{startDate}/{endDate}`
- Dynamic filename generation including organization name and date range

### 4.10.5 Order Detail View

Detailed view of a specific order including order metadata, financial information, and traveler/companion management.

- Display order details using structured Descriptions layout: order number, organization, product, membership, quantity, retail/deal price, currency
- Show traveler table with Primary (green tag) and Companion (blue tag) type indicators
- Allow editing companion details (email and phone) via modal — only when companion email is empty/null
- Validate companion email format and phone number before submission
- Call assignIdentity API on companion update; reload order data on success
- Navigate to traveler profile or user profile on edit action based on user context

### 4.10.6 Manage Organization

Organization settings management with a tabbed interface for address and admin configuration.

- Tabbed interface: Primary Address tab (always visible), Settings tab (visible only for Administrator, CSR, Admin, Travel Manager, Partner roles)
- Plan tab exists in code but is currently disabled/commented out
- Fetch organization data on load using current user's companyID (with localStorage fallback)
- Propagate data changes from child components (PrimaryAddress, AdminSettings) back to parent via callback

- Redirect with error notification on API failure

### 4.10.7 Manage Users

User management interface for viewing, searching, filtering, and managing user status within an organization.

- Tabbed view: All Users, Active Users, Inactive Users
- Column-specific search for: First Name, Last Name, Email, Phone
- Toggle switches for activating/deactivating users with real-time API updates
- Actions per user: Edit (navigate to edit page), View Audit Logs, Add New User
- Server-side pagination with filter and pagination state persisted to Redux
- All user operations scoped to current user's organization

### 4.10.8 Edit / Add User

- Support both Create (id='new') and Edit (id=userId) modes from the same form
- Form fields: First Name, Last Name, Email (disabled in edit mode), Phone, Address, Country, State, Role
- Load role list: internal roles for companyID=1, external roles for all other organizations
- Validate: name regex, email format, phone (min 6 digits, digits/+ only)
- On success: show success notification and navigate back to user list

### 4.10.9 Manage Agents

Agent management dashboard with bulk upload, status management, and role-based filtering.

- Tabbed view: All, Active, Inactive agents
- Table columns: Agent Code, First Name, Last Name, Email, Phone, Status (toggle), Edit action
- Column-specific search with active filter tag display
- Status toggle: if agent has an assigned plan, show confirmation modal before deactivation
- Bulk CSV upload via POST  
`/api/BulkUploadOrganizationAgents/{OrgID}/{sendEmailForNew}`
- Validate CSV file format before upload; show success/error notification and refresh table
- Persist pagination and filter state to Redux across navigation

### 4.10.10 Edit / Add Agent

- Support Create and Edit modes
- Form fields: First Name, Last Name, Email (read-only when editing), Phone, Address, Country, State, Role
- Load external roles for non-internal organizations
- Validate name, email, phone with same patterns as user edit
- On success: notify and navigate back to agent list

### 4.10.11 Manage Travellers

Comprehensive traveler management interface supporting plan assignment, bulk upload, and GuardMe-specific features.

- Table columns: All Membership Numbers, Active Membership (current plan with dates), First Name, Last Name, Email, Phone, Role, Status, Actions
- Tabbed filtering: All, Active, Inactive travelers
- Column-specific search with filter tag display and Redux persistence
- Status toggle: show confirmation if deactivating a traveler with an active plan; unassign plan before deactivation
- Bulk CSV upload with auto-mapping of required fields: FirstName, LastName, Email, ContactNo, DateOfBirth
- Display used/total license counts in a popover header with per-product breakdown
- Organization tree navigation for GuardMe organizations (division drilldown)
- Export traveler list to CSV; download sample CSV for bulk upload template
- Role-based visibility: Division Manager navigation restrictions apply

### 4.10.12 Traveller Plan Management (GuardMe)

Plan assignment drawer for GuardMe organization travelers.

- Open plan assignment drawer for a traveler showing current plan and available licenses
- Fetch licenses by selected product via `OrgAPI.getOrganizationLicensesByProduct`
- Calculate plan end date automatically: 180 days (standard), 100 days, or 365 days (annual) based on product
- Allow plan assignment, start date update, and plan unassignment
- Validate form before calling `updateTravelerPlan` API
- Refresh license counts and traveler list after plan changes

### 4.10.13 Edit / Add Traveler

- Support Create and Edit modes
- Form fields: First Name, Last Name, DOB (no future dates), Email, Phone, Country, Gender, Role (GuardMe only)
- Emergency contact block: First/Last Name, Relationship, Phone
- Conditional role dropdown for GuardMe organizations: only Travel Manager/Membership roles available
- Email field disabled when editing an existing account; separate email update API for email changes
- On success: notify and navigate back to traveler list

### 4.10.14 Division Management

- Display and manage organizational divisions within the hybrid portal
- Support adding new divisions and editing existing division details
- Navigate to division-level traveler listings
- Role-based access: Division Managers see only their division's data

#### 4.10.15 Audit Logs (Activity Log)

Multi-context audit trail viewer that adapts its behavior based on the entity type being audited.

- Supports 8+ entity contexts via the 'for' prop: users, travellers, employees, invoice, organization, b2b2c-organization, reseller, product, pricing
- Table columns: Date/Time (UTC→local), Updated By (email), Activity (field-level change description)
- Activity rendering: 'User Created' for new records, 'User LoggedIn' for timestamp-only changes, 'FieldName changed from A to B' for field changes
- Gender IDs translated to names (Male, Female, Other, Prefer Not To Answer)
- Pricing context: eye icon opens modal with detailed tier price breakdown
- Header and Back button hidden when embedded inside traveler/employee detail pages
- Role-aware Back button navigation: Administrator routes to specific admin pages; others use `history.goBack()`
- Column search provides client-side filtering across all visible log records

#### 4.10.16 Notifications Inbox

Notification management page for viewing, reading, and bulk-managing system notifications.

- Display paginated table: Title (clickable read/unread toggle, bold when unread), Body (rich HTML), Created Date (UTC→local), Actions (envelope icon toggle)
- Render notification body HTML with XSS protection (replace '<script' with 'illegalscript')
- Individual toggle: optimistic local state update (no re-fetch); updates badge count by  $\pm 1$
- Bulk actions via dropdown: Mark All as Read, Mark All as Unread — triggers server re-fetch and badge count refresh
- Badge count integration: `updateNotificationCount(delta)` for individual; `getNotificationsCount(userId)` for bulk
- Restore pagination and filter state on back-navigation
- Filter tags display above table; click to clear individual filters

#### 4.10.17 Customized Email Templates — List

Email template management list for viewing and navigating to edit organization-specific email templates.

- Display table: Title (searchable), Type (formatted from hyphenated key e.g., 'welcome-email' → 'Welcome Email'), Subject (searchable), Action (Edit)
- Three-tier company ID resolution: URL param (admin context) → Redux `userData` → `localStorage` fallback
- Edit navigates to `./edit/{templateId}`
- Read + Edit only — no create or delete from this list

#### 4.10.18 Customized Email Template — Editor

Rich-text email template editor allowing customization of subject, title, and HTML body content.

- Display current template with editable Title, Subject (Ant Design managed fields) and HTML body (JoditEditor WYSIWYG)
- JoditEditor: uses onBlur (not onChange) for performance — captures final content when editor loses focus
- Show available template variables (keywords) as read-only reference (e.g., {{FirstName}}, {{Email}})
- Submit combines: form values (title, subject) + editor content + original metadata (keywords, language, type, organizationId)
- Restore to Default: fetches master template from orgId=1, overrides organizationId to current org, saves — only visible in admin context
- Cancel navigates back without saving
- Visible only for admin context (when URL params.id exists)

#### **4.10.19 Countries Browser (Hybrid)**

- Display countries grouped by region in tabbed interface — same layout as Admin/Countries
- Provide searchable dropdown for quick country access
- Navigate to country risk detail page on selection

#### **4.10.20 Country Risk Intelligence Dashboard**

Comprehensive country risk detail view with multiple tabs for risk data, travel alerts, advisories, city data, and map visualization.

- Display risk level indicator for the selected country
- Integrate Google Maps polygon rendering for country boundary visualization
- Tabbed interface: Map, Report, Cities, Alerts, Advisories, Covid/Region data
- Cities tab and Advisories tab load lazily on first click (not on mount)
- Risk alerts tab: load on mount, filterable by criteria
- Membership/plan expiry gating: users with expired plans see a modal prompt before accessing certain tabs
- Modal behavior varies by user role: Membership users vs Employee users see different prompts
- Full page spinner during initial load; tab content spinner during lazy loads

## 4.11 Admin Reporting & Management Module [NEW — v2.0]

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The Admin Reporting Module provides Administrators and CSRs with comprehensive business intelligence, billing management, traveler tracking reports, and organizational data management tools.

### 4.11.1 Admin Home Dashboard

The main admin landing page providing a high-level business intelligence overview combining financial statistics, system metrics, and quick-access navigation for administrators.

- Display financial KPIs and summary statistics on login
- Provide quick navigation to all major admin sections
- Role-restricted access: Administrator and CSR only

### 4.11.2 Accounts Billing — Invoice Management

Administrative dashboard for managing organization invoices with advanced filtering, editing, and organization hierarchy support.

- Display sortable, paginated invoice table: Organization Name, Division, Billing Company, Membership Number, Plan Name/Quantity, Total Amount (USD), Purchase/Due/Paid Dates, PO Number, Payment Status (Paid/Unpaid tag), Actions
- Column-specific search: Organization Name, Division, Billing Company, Membership Number
- For Administrator/CSR: parent organization dropdown with infinite scroll and search; load child organizations as tree select for filtering
- Edit invoice via modal: PO Number and Payment Status (Paid/Unpaid toggle); disable fields if invoice is already paid
- Validate PO Number as required; payment status changes trigger confirmation modal
- Support URL parameter pre-filtering by organization/division ID
- Navigate to individual invoice detail page via view action

### 4.11.3 Invoice Detail View & PDF Download

Formatted invoice display with client-side PDF generation using @react-pdf/renderer.

- Render a professionally formatted invoice layout: company logo/header (FocusPoint International Inc.), invoice number/date, Bill-To details, payment terms, itemized product table, remittance payment instructions
- Calculate payment terms in days using `Math.ceil(dueDate.diff(createDate, 'days'))`
- Single line-item per invoice (quantity × unit price = total)
- PDF Download button generates identical invoice as downloadable invoice.pdf via client-side PDF rendering
- Pre-resolve all translation labels at mount time for PDF renderer (operates outside React context)

- Display total amount formatted as USD currency
- Note: company banking details (routing number, account, SWIFT) are currently hardcoded — should be moved to configuration

#### **4.11.4 Traveler Itinerary Viewer (Automated — PNR-based)**

Read-only view of a traveler's itinerary fetched by email and PNR ID, showing flights, hotels, cars, and remarks.

- Fetch itinerary using URL params: email and pnrlId via Redux `getItineraryByPnrlIdAndEmail`
- Display sections: Passenger Info, PNR General Info (PNR number, invoice, customer account, org name), Flights, Hotels, Cars, Remarks
- Flight columns: Departure/Arrival airports, times, vendor, flight number, passengers (email tooltip), status (L=LANDED, S=SCHEDULED, A=other)
- Hotel columns: Vendor, name, address, city code, check-in/out dates
- Car columns: Vendor, type, pickup/return locations, dates, confirmation number
- Show original times when they differ from scheduled times
- Sequential counters displayed for flights, hotels, and cars

#### **4.11.5 Manual Itinerary Viewer**

View of manually created traveler itineraries (not PNR-automated), focusing primarily on flights.

- Fetch manual itinerary using pnrlId URL param via Redux `getManualItineraryByPnrlIdAndEmail`
- Display: Passenger Info, PNR General Info (no PNR number), Flights with 'Not Available' fallbacks for missing vendor/flight data
- Note: Hotels and cars are bound in state but not currently rendered (incomplete implementation)

#### **4.11.6 Reports — Active Traveler List**

Administrative report for viewing currently active travelers from both Manual and Magnatech (automated) sources.

- Tabbed interface: Manual (tab 1) and Magnatech (tab 2) traveler lists
- Table columns: Traveler Name, Division, PNR, Departure Date, Routing, View Action; Organization column added for CSR/Administrator roles
- Column-specific search for: Name, Division, Organization, PNR
- CSV download for Manual, Magnatech, or combined reports via Axios blob download with auth headers
- Print functionality via react-print-components
- View action opens traveler itinerary viewer in a new browser window (tab-aware URL construction)
- Tab switch resets all filters and reloads data for the selected source

#### **4.11.7 Reports — Where Are My Travelers**



Traveler location tracking report for a specific date with date picker, dual-source tabbed view, and export capabilities.

- Mandatory DatePicker: form validation ensures date is selected before data loads
- Tabbed interface: Manual and Magnatech sources
- Table columns: Traveler Name, Division, PNR, City, Departure/Arrival Dates, Routing, View Action; Organization column for privileged roles
- Column-specific search with filter tags
- CSV download (Manual, Magnatech, or combined) and print functionality
- Combined CSV download uses a timeout to allow data load before export

#### **4.11.8 Reports — Travelers by Destination & Date**

Advanced traveler tracking report filtering by destination country/city and travel date range.

- Filter by destination country, city, and date range before loading data
- Display matching traveler records with PNR, routing, and itinerary details
- View action navigates to itinerary detail for the selected traveler
- CSV export and print functionality

#### **4.11.9 Reports — Travelers Risk Level**

Report showing traveler risk exposure based on their current travel destinations.

- Display travelers with associated country risk levels for their itineraries
- Support filtering by risk level and date range
- CSV export and print functionality

#### **4.11.10 Reports — Employees on the Same Flight**

Report identifying multiple travelers sharing the same flight — used for risk concentration management.

- Filter by flight number, date, or PNR to identify co-travelers
- Display passenger details, flight info, and organizational context
- CSV export functionality

#### **4.11.11 Reports — Domestic/International & Segment Analysis**

Analytical reports providing breakdown of travel patterns.

- Domestic/International: categorize traveler trips by domestic vs. international routing
- Segment Analysis: display travelers with details like name, division, PNR, departure/arrival dates for route-level analysis
- Both reports support CSV export and filtering

#### **4.11.12 Reports — Airport Reports (Arrival / Departure / Search by Flight)**

Airport-centric reports for viewing travelers at a specific airport on a specific date.

- Airport Report: query travelers associated with a selected airport on a specific date
- Arrival Report: travelers arriving at a selected airport on a specific date, categorized by arrival status
- Departure Report: travelers departing from a selected airport on a specific date
- Search by Flight: query travelers on a specific flight number
- All airport reports support: date picker input, paginated results, CSV export, and print functionality
- Search result sub-pages display the filtered traveler list

#### **4.11.13 Reports — Active Hotels & Active Car Rentals**

- Active Hotel Report: list of travelers currently staying in hotels with hotel name, check-in/out dates, and location
- Active Car Rental Report: list of travelers with active car rentals showing rental company, vehicle type, and dates
- Both reports support CSV export and print functionality

#### **4.11.14 Reports — Travelers by Active Alert**

Report showing travelers currently in regions affected by active risk alerts.

- Display travelers impacted by each active risk alert with traveler details and alert information
- Sub-page (ViewTravelersByActiveAlertReport) shows the specific traveler list for a selected alert
- CSV export and print functionality

#### **4.11.15 Reports — Report Hub**

Central navigation page providing access to all available report types for the user's role.

- Display all available report links based on user role and permissions
- Quick navigation to: Active Travelers, Where Are My Travelers, Travelers by Destination/Date, Risk Level, Same Flight, Domestic/International, Segment Analysis, Airport/Arrival/Departure, Active Hotels, Active Cars, Active Alert reports

#### **4.11.16 B2C Traveler Management**

Management interface for direct B2C consumers who purchased plans individually (not through an organization).

- Display paginated, searchable table of all B2C travelers: Membership Number(s), First Name, Last Name, Email (userName), Phone
- Membership Number column renders multiple numbers on separate lines (comma-split membershipNumbers)
- Actions: Edit (navigate to traveler profile), Resend Membership Info (sends membership email via b2cActions.resendMembershipInfo)
- All five data columns support column-specific search with Redux filter persistence
- No organization ID filter — B2C travelers are platform-wide users

- Clear existing data array immediately on table reload for instant visual feedback

#### **4.11.17 B2C Traveler Profile (Hub View)**

Comprehensive detail view for a B2C traveler's profile organized in a tabbed interface.

- Tabbed interface covering: Personal Details, Plan History, Activity Logs, Map Location, Check-In History, and Assist Requests
- Notification drawer for sending push notifications directly to a traveler
- Organizes multiple child components into a single management hub

#### **4.11.18 Organization License Management**

Paginated, searchable view of all plan licenses for a specific organization with aggregate usage statistics.

- Table columns: License Number, Product, Company Name, Division Name, Assigned User, Assigned User Email, Status (Active/Inactive/Expired), Actions (View)
- Aggregate header: Total Assigned / Total Available licenses with per-product breakdown in hover popover
- Column-specific search for all data columns with Redux filter persistence
- Three independent operations on mount: fetch org name, restore pagination, fetch license usage counts
- Role-aware Back button navigation: different path depths for Division Manager vs. Travel Manager vs. Admin/CSR
- Note: Component exists in three identical copies at different route paths — consolidation recommended

#### **4.11.19 License History (Audit Log)**

Chronological read-only audit log of all events for a specific license.

- Display license history: Company Name, Division Name, Username, User Email, Event Type, Date/Time (MM/DD/YYYY HH:mm)
- Event types: Assigned, Unassigned, Transferred
- Column search for: Company Name, Division Name, Username (Email search is currently disabled in UI)
- Page title shows: '(MEMBERSHIP\_NUMBER-LICENSE\_ID)' for unique identification
- Always starts fresh — no pagination state restoration between visits

#### **4.11.20 Organization Order History**

Read-only order detail view for organizational plan purchases — informational only, no actions.

- Display order details: Company Name, Division Name, Bill To, Product Name, Membership Number, Number of Plans, Retail Price, Deal Price, Coverage Start Date, Coverage End Date
- Formatted currency using Intl.NumberFormat USD
- No action buttons or modals — purely informational

#### **4.11.21 Admin — User Management (Internal)**

- Internal admin user management: create, view, and manage system-level users
- Search users by various criteria; edit user roles and status
- View user audit logs from the user management list

#### **4.11.22 Admin — Products Management**

- Manage travel assistance products (plan types)
- View product details and change history via product audit log
- Product listing with pagination and search

#### **4.11.23 Admin — Roles Management**

- View and manage system roles
- Edit role permissions and assignments
- Roles drive UI feature visibility and API access control throughout the platform

#### **4.11.24 Admin — COVID / Region Data**

- Manage COVID and regional health advisory data for countries
- Displayed in the Country Risk Intelligence Dashboard as a separate tab

## 5. Non-Functional Requirements

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### 5.1 Performance

- Page load times should not exceed 3 seconds on standard broadband connections
- The pricing engine must recalculate totals in real time without perceptible delay
- Paginated tables should load within 2 seconds per page change
- Map dashboard initial load should complete within 5 seconds including all marker data
- CSV exports should complete within 10 seconds for typical dataset sizes

### 5.2 Security

- All API calls must use authentication tokens (JWT or equivalent) passed via Authorization headers
- Sensitive data (passwords, tokens, banking details) must never be stored in frontend code
- Password policy: minimum 8 characters, uppercase, lowercase, numeric, special character required
- Stripe PCI compliance: card details must never be stored by the application
- Role-based access control (RBAC) must restrict features to appropriate user roles
- HTML content rendered via `dangerouslySetInnerHTML` must be sanitized (replace '<script' with 'illegalscript' at minimum)

### 5.3 Usability

- Forms must provide real-time inline validation feedback
- Loading spinners must be displayed during all async operations
- All monetary values must be consistently formatted (e.g., \$123.45 USD)
- The application must be responsive across desktop and mobile screen sizes
- Date inputs must use locale-aware pickers with future-date restrictions where applicable
- All UI text must support internationalization via `IntlMessages`

### 5.4 Reliability & Availability

- The system must handle API failures gracefully with user-facing error notifications
- Form state must be preserved across page refreshes using `localStorage` where feasible
- The B2C purchase wizard must be re-entrant: users who refresh mid-flow should not lose progress
- Pagination and filter state must be persisted to `Redux` to survive navigation within a session

### 5.5 Scalability

- The system must support multi-tenant operation with isolated theming and data per reseller
- Pricing, country-tier, and role lookup tables must support updates without frontend deployments

## 5.6 Integrations

| Integration          | Purpose                                                                       | Notes                                                    |
|----------------------|-------------------------------------------------------------------------------|----------------------------------------------------------|
| Stripe               | Payment processing for B2C and plan upgrade flows                             | StripeProvider component; API key from config            |
| Cigna                | Employee plan provisioning and access management                              | One-shot approve/reject URLs; CreateUserAccess API       |
| Google Maps          | Interactive travel map dashboard with country polygons, markers, flight paths | Used in Hybrid Dashboard — itinerary-details-map.jsx     |
| RiskLine API         | Country risk level data and alert categories for map and reports              | Risk alerts filterable by country, level, time, category |
| Google Analytics     | User behavior tracking in production                                          | ReactGA — measurement ID G-N02B8BDNND                    |
| Microsoft Clarity    | Session recording and heatmaps in production                                  | Tracking ID: offz2ddba7                                  |
| @react-pdf/renderer  | Client-side PDF generation for invoice download                               | PdfDocument component renders invoice as PDF             |
| Jodit WYSIWYG Editor | Rich HTML editing for email template customization                            | jodit-react; uses onBlur for performance                 |
| CAP Travel Website   | External quote builder and plan information                                   | Redirect via window.location.href                        |

## 6. Key Data Entities

### 6.1 User / Traveler

| Field                 | Description                                                |
|-----------------------|------------------------------------------------------------|
| userID                | Unique user identifier                                     |
| firstName, lastName   | User's full name                                           |
| email / userName      | Login and communication email (used as username)           |
| dateOfBirth           | Required for age validation (18+)                          |
| contactNo / phone     | Required for plan purchase (min 6 digits)                  |
| countryId             | Home country (required for purchase)                       |
| agentCode             | Used to filter orders for agents                           |
| companyID / orgId     | Organization association                                   |
| accountConfirmed      | Whether the account has been confirmed via password change |
| isReceiveTravelAlerts | Opt-in for travel alert notifications                      |
| gender                | Required for traveler registration (GuardMe)               |
| roleId                | Role within the organization (GuardMe organizations)       |

### 6.2 Order / Plan

| Field                               | Description                                                          |
|-------------------------------------|----------------------------------------------------------------------|
| orderId                             | Unique order identifier                                              |
| productName                         | Name of the purchased travel plan                                    |
| membershipNumber(s)                 | Membership identifier(s) linked to the order (B2C may have multiple) |
| orderNumber                         | External order reference number                                      |
| agentTrackingNumber                 | Tracking number for agent-sold orders                                |
| insertDate                          | Order creation timestamp                                             |
| totalOrderRetailAmount              | Full retail price before discounts                                   |
| totalOrderAmount                    | Discounted/deal price actually charged                               |
| currency                            | Currency code for the order (USD, CAD)                               |
| coverageStartDate / coverageEndDate | Coverage period for the plan                                         |
| orgId                               | Organization under which the order was placed                        |

## 6.3 Invoice

| Field                             | Description                                   |
|-----------------------------------|-----------------------------------------------|
| invoiceId                         | Unique invoice identifier                     |
| organizationName / divisionName   | Org and division context                      |
| billingOrganization               | Billing entity name                           |
| productName / productQty          | Product and quantity purchased                |
| totalAmount                       | Total invoice value (USD)                     |
| purchaseDate / dueDate / paidDate | Key dates for payment tracking                |
| poNumber                          | Purchase Order number (editable by Admin/CSR) |
| paymentStatus                     | Paid / Unpaid                                 |

## 6.4 License

| Field                            | Description                                                     |
|----------------------------------|-----------------------------------------------------------------|
| licenseNumber                    | Unique license identifier                                       |
| product                          | Coverage product type assigned to this license                  |
| organizationName / divisionName  | Organizational context                                          |
| assignedUser / assignedUserEmail | Current license holder                                          |
| status                           | Active / Inactive / Expired                                     |
| history                          | Chronological log of Assigned / Unassigned / Transferred events |

## 6.5 B2C Wizard State (Redux)

| Field              | Description                                                                       |
|--------------------|-----------------------------------------------------------------------------------|
| planType           | Selected plan tier (product object)                                               |
| planTerm           | Term: dates, day package (dp), countries, tier value, total price, CAD price      |
| personalDetails    | Array of traveler objects (name, DOB, email, country, product, emergency contact) |
| billingInformation | Payment method and billing details                                                |
| couponInfo         | Applied promo code details (type, discount amount)                                |



| Field                 | Description                         |
|-----------------------|-------------------------------------|
| step                  | Current wizard step (1-5)           |
| isAnnual              | Whether an annual plan was selected |
| totalPrice / cadPrice | Calculated prices in USD and CAD    |

## 7. Identified API Endpoints

The following API endpoints were identified from frontend API calls across all code review iterations. All endpoints use the Config.API1 base URL and require authentication headers unless otherwise noted.

| Endpoint / Action                                              | Method       | Purpose                                              |
|----------------------------------------------------------------|--------------|------------------------------------------------------|
| signIn / login                                                 | POST         | Authenticate user                                    |
| forgotPassword                                                 | POST         | Send password reset email                            |
| resetPassword                                                  | POST         | Reset password using token                           |
| updatePassword                                                 | PUT          | Change authenticated user's password                 |
| myAccount / getTraveller                                       | GET          | Current user or specific traveler profile            |
| updateMyAccount                                                | PUT          | Update user profile                                  |
| validateTravellerEmail                                         | POST         | Check if email already exists                        |
| /api/ExportAgentOrders/{orgId}/{agentId}/{startDate}/{endDate} | GET          | Export agent orders as CSV blob                      |
| getAgentOrders / getAgentListing                               | POST/<br>GET | Fetch paginated agent order/listing                  |
| updateEmployeeStatus                                           | PUT          | Activate / deactivate an agent or user               |
| updateUser                                                     | PUT          | Add or edit agent/user profile                       |
| /api/BulkUploadOrganizationAgents/{OrgID}/{sendEmailForNew}    | POST         | Bulk upload agents from CSV                          |
| getTravellersListing / getB2CTravellersListing                 | GET          | Fetch paginated traveler list (org or B2C)           |
| addTraveller / updateTravellerData                             | POST/<br>PUT | Create or update traveler profile                    |
| updatetravelerEmail                                            | PUT          | Update traveler email (separate from profile update) |
| resendMembershipInfo                                           | POST         | Resend membership information email to traveler      |
| /api/BulkUploadTravellers                                      | POST         | Bulk upload traveler CSV with field mapping          |
| getGuardMeProducts / getLicenseCount                           | GET          | Fetch GuardMe products and license usage             |
| OrgAPI.getOrganizationLicensesByProduct                        | GET          | Fetch available licenses for a specific product      |
| updateTravelerPlan / guardMeAssignLicense                      | PUT/<br>POST | Assign or update traveler plan                       |

| Endpoint / Action                                                     | Method    | Purpose                                                |
|-----------------------------------------------------------------------|-----------|--------------------------------------------------------|
| getOrganizationLicenses / getLicenseLogs                              | GET       | Fetch organization licenses or license audit log       |
| getResellerOrders                                                     | GET       | Fetch paginated reseller/org order history             |
| /api/ExportOrders/{orgId}/{startDate}/{endDate}                       | GET       | Export reseller orders as CSV blob                     |
| getOrder                                                              | GET       | Fetch single order detail                              |
| assignIdentity                                                        | POST      | Assign email/phone to a companion traveler on an order |
| getInvoice / updateInvoice                                            | GET/PUT   | Fetch or update invoice (PO number, payment status)    |
| getOrganizationId                                                     | GET       | Fetch organization details by ID                       |
| getUserListing                                                        | GET       | Fetch paginated user list for an organization          |
| getCountriesListByRegion / getCountriesList                           | GET       | Fetch countries grouped by region or flat list         |
| /images/{code}.png                                                    | GET       | Country flag image by ISO code                         |
| getCountriesListWithRiskLevel                                         | GET       | Countries with risk level data for map coloring        |
| getRiskLineCategories / getRiskLineAlerts                             | GET       | Risk alert categories and alert data from RiskLine API |
| getItineraryByPnrIdAndEmail                                           | GET       | Fetch traveler itinerary by PNR and email              |
| getManualItineraryByPnrIdAndEmail                                     | GET       | Fetch manual itinerary by PNR                          |
| getManualActiveTravelers / getActiveTravelers                         | GET       | Active traveler list (manual or Magnatech)             |
| getWhereAreMyTravelers / getManualWhereAreMyTravelers                 | GET       | Travelers by location on a specific date               |
| getNotificationsList / updateNotificationStatus                       | GET/PUT   | Fetch notifications; toggle read/unread                |
| getNotificationsCount / updateNotificationCount                       | GET/PATCH | Fetch or adjust notification badge count               |
| getEmailTemplates / getEmailTemplate                                  | GET       | List or fetch single email template                    |
| updateEmailTemplate                                                   | PUT       | Save customized email template                         |
| getUserLog / getOrgLogs / getProdLogs / getPricingLog / getInvoiceLog | GET       | Fetch audit logs by entity type                        |
| /api/approvecreateuser/{id}                                           | GET       | Approve Cigna user access (token-based, no auth)       |
| /api/rejectcreateuser/{id}                                            | GET       | Reject Cigna user access (token-based, no auth)        |

| Endpoint / Action                                              | Metod | Purpose                                             |
|----------------------------------------------------------------|-------|-----------------------------------------------------|
| CreateUserAccess                                               | POST  | Create Cigna user access record                     |
| placeOrder                                                     | POST  | Submit final B2C order                              |
| getProductPricingAndInfo /<br>getResellerProductPricingAndInfo | GET   | Product pricing tiers and org info                  |
| getAppliedPromo                                                | GET   | Fetch applied promo/coupon details                  |
| OrgAPI.B2BEmployeeUpgrade                                      | POST  | Check plan upgrade eligibility by email             |
| CheckEmployeeChild / EmployeeExits                             | POST  | Validate child/parent email in family purchase flow |

## 8. Assumptions, Constraints & Risks

### 8.1 Assumptions

- The backend (.NET and Java) services expose RESTful APIs consumed by the React frontend as analyzed
- Redux state is the single source of truth for multi-step wizard data; backend stores persist on final order submission
- Stripe handles all payment card processing; the application does not store raw card data
- The CAD conversion rate (1.3644) is hardcoded; a dynamic FX rate API may be required for production
- Company banking details in the invoice component (routing number, account, SWIFT) should be moved to secure backend configuration
- The OrganizationLicenseList component exists as three identical copies at different routes — consolidation to a single shared component is assumed as future tech debt
- GuardMe-specific features (plan assignment, bulk upload, org tree) are conditional on isThisGuardMeOrg flag returned by OrgAPI.getOrganization

### 8.2 Constraints

- Backend could not be run locally during analysis; all backend requirements are inferred from frontend code
- Some UI features are partially implemented: date range filter on agent order history is commented out; SMS alerts toggle in Cigna form is not wired; Manual Itinerary Viewer does not render hotels/cars
- The WhereareTravelersResult component uses hardcoded sample data — not connected to real API (prototype/demo)
- Family purchase coverage end date is hardcoded; requires backend-driven configuration
- The DOB age check contains a potential bug (Date.getDay() used instead of Date.getDate()) that must be reviewed
- OrganizationLicenseList triplicate files represent ongoing technical debt

### 8.3 Risks

| Risk                                                            | Likelihood | Impact | Mitigation                                                                     |
|-----------------------------------------------------------------|------------|--------|--------------------------------------------------------------------------------|
| Backend API contracts differ from frontend expectations         | Medium     | High   | Conduct API contract review with backend team; establish OpenAPI documentation |
| Hardcoded banking details in invoice cause update failure       | Medium     | High   | Move to backend config or environment variable; never in source code           |
| DOB validation bug (getDay vs getDate) allows underage purchase | Low        | High   | Fix: replace Date.getDay() with Date.getDate() in age check logic              |
| Hardcoded CAD exchange rate (1.3644) causes pricing errors      | High       | Medium | Replace with dynamic FX rate API integration                                   |

| Risk                                                                | Likelihood | Impact | Mitigation                                                                      |
|---------------------------------------------------------------------|------------|--------|---------------------------------------------------------------------------------|
| Three identical OrganizationLicenseList components cause divergence | High       | Medium | Consolidate to single shared component with route-context props                 |
| Google Maps API key not managed per environment                     | Low        | High   | Ensure Maps API key is environment-specific and not committed to source control |
| Session loss during B2C checkout on page refresh                    | Medium     | Medium | localStorage persistence is implemented; verify completeness across all steps   |
| RiskLine API unavailability affects map dashboard                   | Low        | Medium | Implement graceful degradation — show map without risk data if API unavailable  |

## 9. Glossary

| Term              | Definition                                                                                                  |
|-------------------|-------------------------------------------------------------------------------------------------------------|
| B2C               | Business-to-Consumer: Direct consumer purchase flow via the CAP portal                                      |
| B2B               | Business-to-Business: Agent-driven sales model                                                              |
| B2B2C             | Business-to-Business-to-Consumer: Reseller/white-label portal model                                         |
| CAP               | Care and Assistance Plus — the travel assistance plan product brand                                         |
| GuardMe           | A specific organization type within B2B2C with enhanced plan assignment and license management features     |
| Tier              | Country/risk classification level used to determine plan pricing                                            |
| Day Package (dp)  | Coverage duration in days (e.g., 8, 15, 30, 60, 90, 120, 365)                                               |
| PNR               | Passenger Name Record — unique booking reference used to identify itineraries                               |
| Redux             | JavaScript state management library used to manage wizard and session state                                 |
| Wizard            | The multi-step B2C purchase flow (Steps 1–5)                                                                |
| orgId / companyID | Organization/company identifier used to scope data                                                          |
| agentCode         | Unique identifier for an agent within their organization                                                    |
| Stripe            | Third-party payment gateway used for credit/debit card processing                                           |
| Cigna             | Healthcare/insurance partner with a dedicated user provisioning integration                                 |
| RiskLine          | Third-party risk intelligence provider supplying country risk levels and alerts                             |
| JWT               | JSON Web Token — presumed authentication mechanism for API calls                                            |
| PriceCalculation  | Frontend component that computes real-time plan pricing                                                     |
| placeOrder        | API action that submits the final confirmed order to the backend                                            |
| IntlMessages      | Internationalization service used throughout the application for translated UI text                         |
| Jodit             | Rich-text WYSIWYG HTML editor used in the Email Template Editor                                             |
| GuardMe Org       | Organization with isThisGuardMeOrg=true enabling plan assignment, license management, and org tree features |

## 10. Appendix: Analyzed Components

All components analyzed across three code review iterations (original + Code\_1 + Code\_2):

| Component / File             | Route                        | Description                                          |
|------------------------------|------------------------------|------------------------------------------------------|
| AgentOrderHistory            | /agent/order-history         | Agent order table with search, export, pagination    |
| Login                        | /login                       | User authentication form                             |
| Countries (Admin)            | /admin/countries             | Admin country management with regional tabs          |
| UserProfile (Agent)          | /agent/my-profile            | Agent profile view and edit form                     |
| UserChangePassword (Agent)   | /agent/change-password       | Secure password change with real-time validation     |
| ForgotPassword               | /forgot-password             | Password reset request form                          |
| ResetPassword                | /reset-password              | Token-based password reset form                      |
| reloadId.js                  | /cap/dashboard               | Route alias/redirect handler with profile validation |
| plan-type.jsx                | /cap/purchase                | B2C Step 1: Plan type selection                      |
| plan-term                    | /cap/plan-term               | B2C Step 2: Plan term selection                      |
| personal-details.jsx (B2C)   | /cap/personal-details        | B2C Step 3: Traveler registration form               |
| billing-information.jsx      | /cap/billing-information     | B2C Step 4: Billing with Stripe payment              |
| purchase-summary.jsx         | /cap/purchase-summary        | B2C Step 5: Order review and confirmation            |
| personal-details.jsx (Order) | /cap/order                   | Quote-initiated Step 3 with URL param decoding       |
| billing-summary.jsx          | /cap/billing-summary         | Quote-initiated Step 4 with Stripe                   |
| price-calculation.jsx        | Shared B2C                   | Real-time pricing engine with tier and promo logic   |
| ManageCigna / index.js       | /cigna                       | Cigna user access provisioning form                  |
| accept.js                    | /cigna/approvecreateuser/:id | Cigna one-shot user activation page                  |
| reject.js                    | /cigna/rejectcreateuser/:id  | Cigna one-shot user rejection page                   |
| ManagePlanUpgrade            | /plan-upgrade                | Plan upgrade eligibility check and Stripe payment    |
| purchase-plan.jsx            | /plan-upgrade/purchase-plan  | Plan upgrade placeholder/wrapper                     |



| Component / File             | Route                                 | Description                                              |
|------------------------------|---------------------------------------|----------------------------------------------------------|
| childParentPurchase          | /family-purchase/:id                  | Family plan purchase form with multi-traveler support    |
| thank-you.jsx (family)       | /family-purchase/:id/thank-you        | Family purchase confirmation page                        |
| B2B2C / index.js             | /partner                              | B2B2C reseller portal with dynamic theming and analytics |
| itinerary-details-map.jsx    | /hybrid                               | Travel map dashboard (Google Maps, risk, markers)        |
| user-profile.jsx (hybrid)    | /hybrid/my-profile                    | Hybrid user profile with emergency contact               |
| change-password.jsx (hybrid) | /hybrid/my-profile/change-password    | Hybrid change password                                   |
| order-history.jsx (hybrid)   | /hybrid/manage-orders                 | Reseller order history with USD/CAD commission           |
| view.jsx (order)             | /hybrid/manage-orders/view/:orderId   | Order detail with companion editing                      |
| edit-organization.jsx        | /hybrid/manage-organization           | Org settings: address + admin tabs                       |
| ManageUsers (hybrid)         | /hybrid/manage-users                  | User list: tabs, search, status toggle, audit log        |
| UserEdit                     | /hybrid/manage-users/:id              | Create/edit user form with role selection                |
| ManageAgents (hybrid)        | /hybrid/manage-agents                 | Agent list with bulk CSV upload                          |
| EditAgent                    | /hybrid/manage-agents/:id             | Create/edit agent profile                                |
| ManageTravelers (hybrid)     | /hybrid/manage-travellers/:orgId/list | Traveler list with plan assignment drawer                |
| EditTraveler                 | /hybrid/manage-travelers/.../edit/:id | Create/edit traveler with GuardMe role                   |
| Division management          | /hybrid/manage-division               | Division add/edit/traveler views                         |
| ActivityLog                  | /hybrid/audit-logs                    | Multi-context audit trail viewer                         |
| Notifications                | /hybrid/notifications                 | Notification inbox with read/unread/bulk actions         |
| CustomizeEmailsList          | /hybrid/customizedEmailList           | Email template list                                      |
| EditCustomizeEmail           | /hybrid/customizedEmail/edit          | Jodit WYSIWYG email template editor                      |
| Countries (hybrid)           | /hybrid/countries                     | Country browser with regional tabs                       |

| Component / File                         | Route                                               | Description                                      |
|------------------------------------------|-----------------------------------------------------|--------------------------------------------------|
| CountriesInfo                            | /hybrid/countriesInfo                               | Country risk intelligence dashboard (map + tabs) |
| accounts-billing/index.jsx               | /user/AccountBilling                                | Invoice management with org hierarchy filter     |
| view-invoice.jsx                         | /user/accounts-billing/or<br>ganization-invoice/:id | Invoice detail view with PDF download            |
| traveler-itinerary-viewer.jsx            | /user/TravelerItineraryVi<br>ewer                   | Automated itinerary view (PNR + email)           |
| traveler-manualitinerary-viewer.js       | /user/TravelerManualItin<br>eraryViewer             | Manual itinerary view                            |
| active-traveler-list                     | /user/reports/active-trav<br>eler-list              | Active travelers report (Manual + Magnatech)     |
| where-my-travelers                       | /users/reports/Where-ar<br>e-my-travelers           | Where are my travelers (date-based)              |
| TravelersbyDestDate                      | /user/TravelersbyDestD<br>ate                       | Travelers by destination and date                |
| TravelersRiskLevel                       | /user/TravelersRiskLeve<br>l                        | Travelers risk level report                      |
| EmployeesonthesameFlight                 | /user/Employeesonthes<br>ameFlight                  | Same-flight traveler report                      |
| DomesticInternational                    | /user/DomesticInternati<br>onal                     | Domestic vs. international travel report         |
| SegmentAnalysis                          | /user/SegmentAnalysis                               | Segment-level travel analysis                    |
| Airport/Arrival/Departure/Flight reports | /user/reports/airport-rep<br>ort (+ others)         | Airport-centric traveler reports                 |
| Active Hotel / Active Car reports        | /user/reports/active-hote<br>l + active-car-report  | Active accommodation/rental reports              |
| Travelers by Active Alert                | /user/reports/travelers-b<br>y-active-alert         | Alert-impacted traveler report                   |
| ManageTravelers (B2C)                    | /user/travelers                                     | B2C traveler listing with resend membership      |
| ViewTravelers                            | /user/travelers/:id                                 | B2C traveler profile hub (tabbed)                |
| OrganizationInvoice                      | /user/OrganizationInvoic<br>e                       | Invoice detail + PDF for org context             |
| ViewOrganizationOrder                    | /user/ViewOrganization<br>Order                     | Read-only org order detail                       |
| OrganizationLicenseList                  | /user/OrganizationLicen<br>seList (x3)              | License list with usage stats (3 route contexts) |

| Component / File        | Route                                                | Description                           |
|-------------------------|------------------------------------------------------|---------------------------------------|
| EditOrganizationLicense | /user/EditOrganizationLicense                        | License history audit log (read-only) |
| AdminHome               | /admin                                               | Admin dashboard home with KPIs        |
| Admin COVID             | /admin/covid                                         | COVID/region data management          |
| Admin Users             | /admin/musers +<br>/admin/users +<br>/admin/useredit | Internal admin user management        |
| Admin Products          | /admin/products +<br>/admin/products/:id             | Product management with audit log     |
| Admin Roles             | /admin/roles +<br>/admin/editrole                    | Role management                       |
| Admin Search            | /admin/search                                        | Global admin search                   |